

Extra homework 3

1. Does there exist a bijective map $\pi : \mathbb{N} \rightarrow \mathbb{N}$ such that $\sum_{n=1}^{\infty} \frac{\pi(n)}{n^2} < \infty$?
2. Let (x_n) be a decreasing sequence of positive numbers. Show that if $\sum_{n \geq 1} x_n$ converges, then

$$\lim_{n \rightarrow \infty} nx_n = 0.$$

3. Start with the interval $[0, 1]$ and remove the middle third $(1/3, 2/3)$; then remove the middle third of each of the remaining intervals, and so on. Denote by $\mathcal{C}$ set of points that are left – this is known as the Cantor set.

- What is the total length of the removed intervals?
- Prove that $x \in \mathcal{C}$ if and only if $x = \sum_{n \geq 1} a_n 3^{-n}$, with $a_n \in \{0, 2\}$.

In other words, $x \in \mathcal{C}$ iff it can be written in base 3 using only 0's and 2's.

- By constructing a surjective function from $\mathcal{C}$ to $[0, 1]$, prove that $\mathcal{C}$ is uncountable.

These questions are extra. You will get bonus points for solving them.